Basic details of team and Problem statement

SMART INDIA HACKATHON 2024

Team Name	**Team Hydro**
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Theme Name	Clean & Green Technology
PS Code	1574
Problem statement Title	Hydrogen Production Facility
Ministry	Ministry of Power

Introduction

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Problem Statement :

- Build a detailed concept for an industrial scale **green hydrogen** (> 50 TPD) production facility, with a levelized cost of hydrogen (LCOH) below $2 USD per kg.

The Project details includes inputs on :

Sizing of production capacity. Production technology, useful life, replacement costs etc.

Proposed RTC-RE model. Proposed sourcing of Feedstock Expected plant CUF.

Entire Financial Model with detailed breakup for CAPEX & OPEX. Storage technologies used (with compliance of safety standards) and their cost.

Approaches used to lower levelized cost during project modelling. Policy and Regulatory support required. Scope for further cost optimisation through economies of scale.

Solution :

Approach

- A hydrogen facility is proposed with industrial scale capacity
- **Solar and Wind**; hybrid farm is to meet the load requirements of Electrolyzer and storage
- **Low-head water turbine** is used to compensate the intermittency of HRES.
- **Ramagundam**; an industrial city in the state of Telangana is chosen for the site

Unique Concept

- **Solid oxide electrolysis cell (SOEC)** electrolyzer units are used which has output purity of hydrogen more than **95%.**
- **Solhyd**; a panel that directly converts sunlight and moisture in atmosphere into green hydrogen is added to system.
- To reduce the **CAPEX** and **OPEX** of the system **green ammonia** storage is proposed.

Research Involved

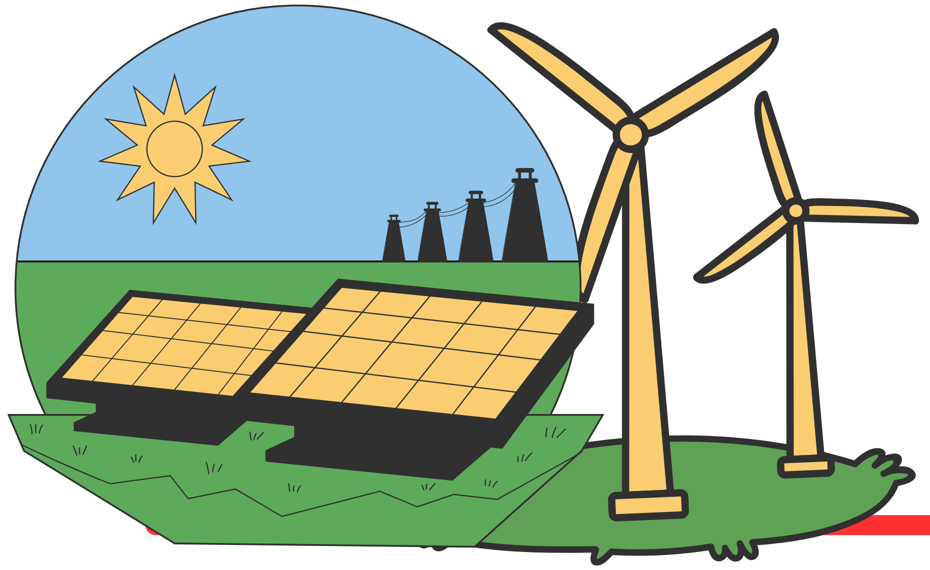
- **Sizing and commissioning parameter** are to be optimized and these details are researched upon.
- The hydrogen sources are diversified and integration of hydrogen panels is utilized.
- **Ammonia storage** and transport economic feasibility is studied
- **Catalysis** is used to convert this green ammonia again into hydrogen to be transported via **pipelines.**

Addressing Problems

- Issues such as **high LCOH** is addressed by assessing alternative production and storage methods.
- The issue of intermittency is catered by the addition of low head water turbine and **hydrogen panels.**
- The storage of ammonia being cheaper is put into application and is powered by the low head turbines and the fuel cell energy from solhyd

Technical Approach : *Proposed Hydrogen Facility*

RE System	Power Produced
Solar PV	**130MW**
Wind Turbines	**400 MW**
Total Power	**530 MW**

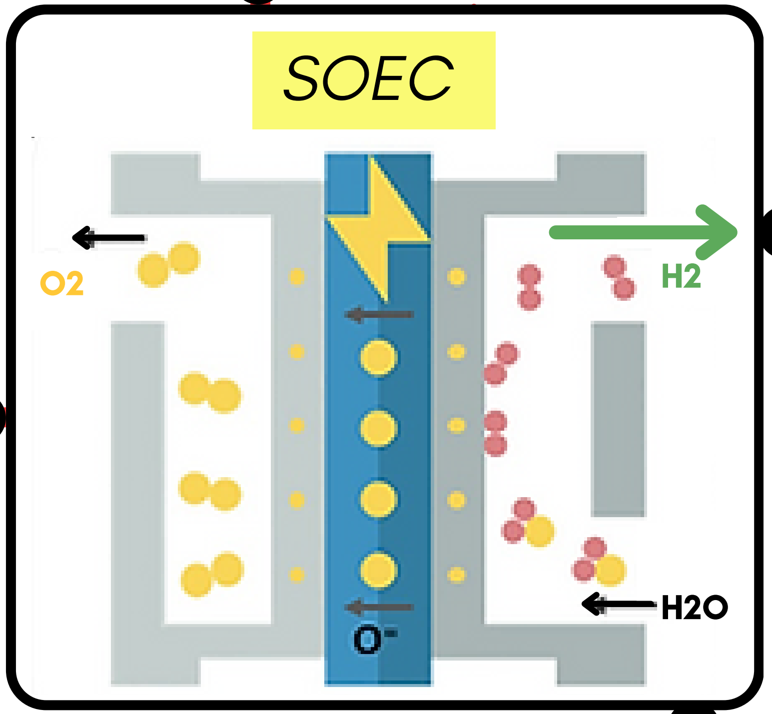


Sustainable **Energy** Generation

Solhyd panel : Solar to H2

For 50TPD;Load Requirements

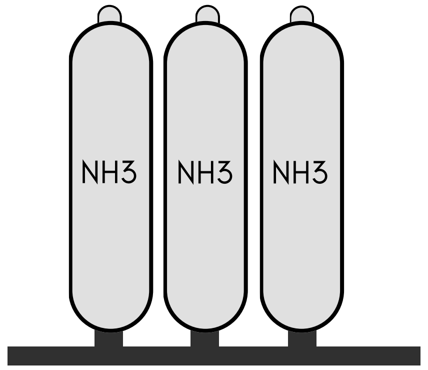
System	Load (MW)
SOEC Electrolyzer	80(10 per stack)
Compressor	160
Storage Units	700



Hydrogen

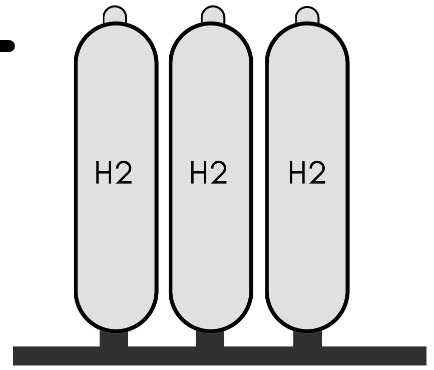
Haber Bosch

Ammonia



Ammonia storage and lifetime is 300 times cheaper than that of compressed green hydrogen

Cracking compression

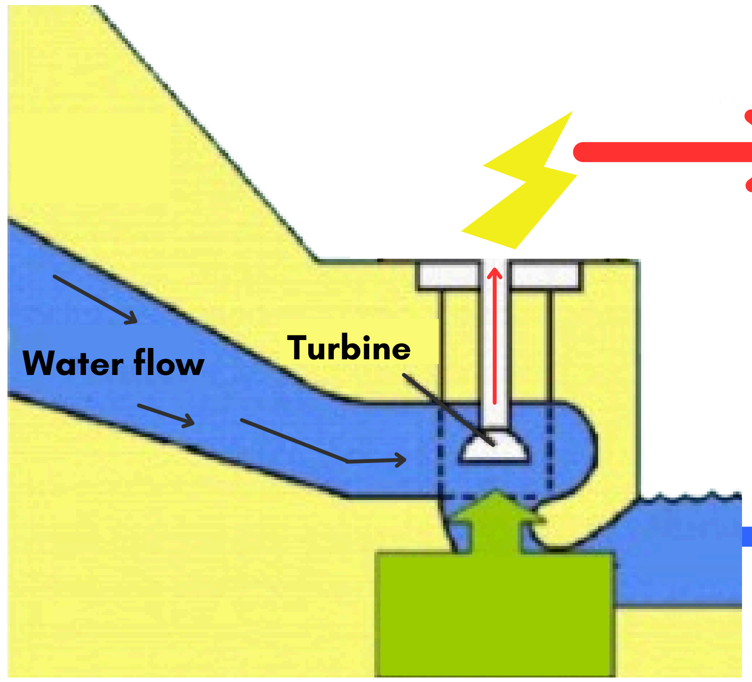


End use (34-44%)

Water supply through **Dams** and **Rivers**

Water Inlet	WR*
Direct water inlet	3.5
Through turbines	2

*Water requirement (Mn liters/day)

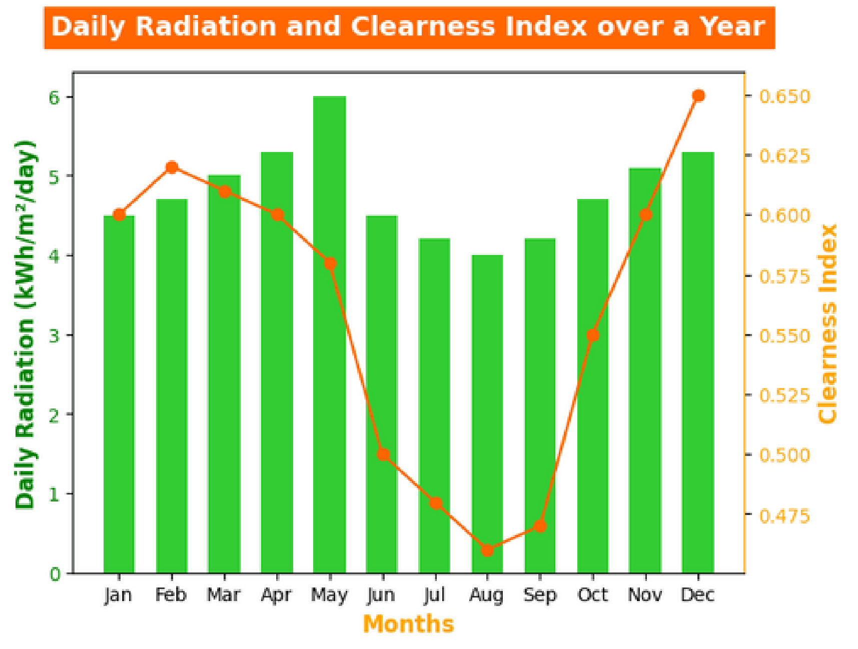


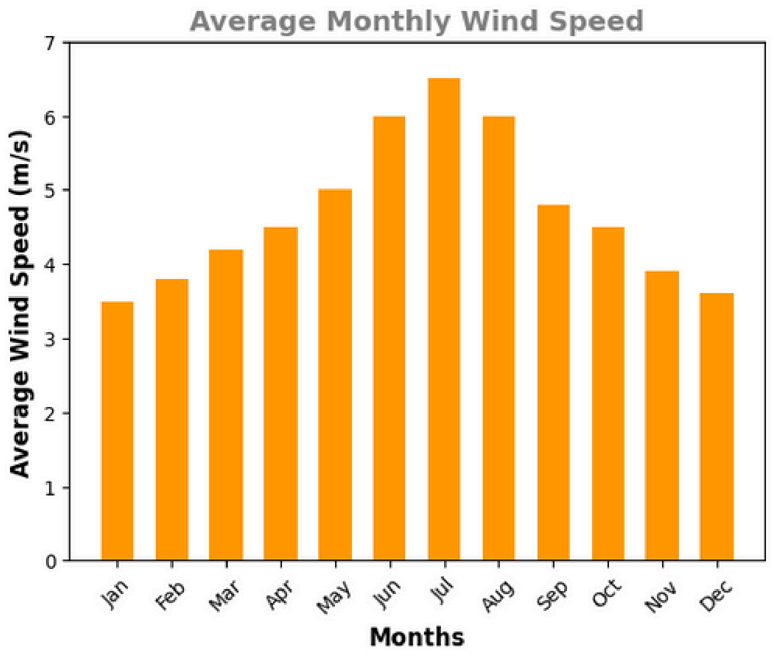
Water flows through low head **water-turbine** and generate electricity.

Direct water supply

Feasilbility and Viabilty

Solar Irradiance and Wind Potential of Site





Site specific; Ramagundam's RE profile

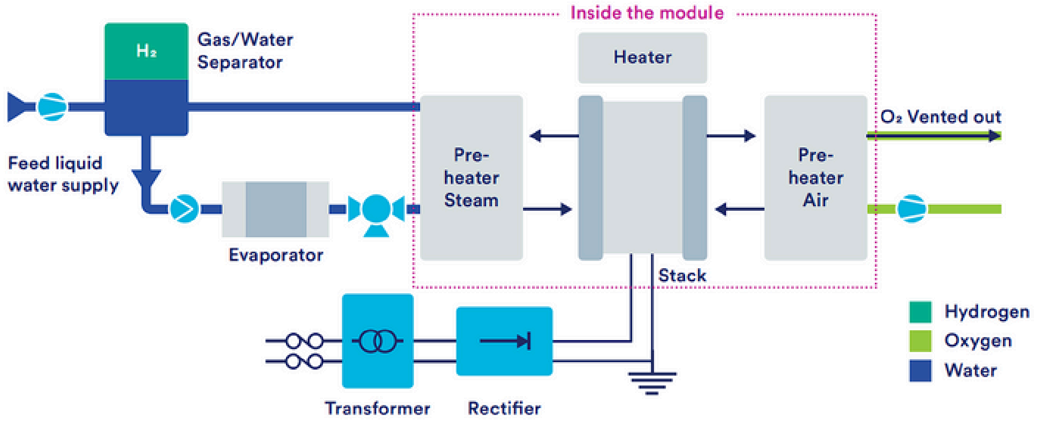
SOEC Working :

Chemical reactions involved are as follows:

Anode Reaction: O2-→O2+2e-

Cathode Reaction: H2O+2e-→H2+O2-

Net Cell Reaction: H2O→H2+O2



Feedstock Availability:

- As per the Environment Sustainability report 2022 by Ministry of Coal, SCCL Ramagundam has 933 ha (2300 acres) of land under lease of which 13% is the active mining area and 67% of the land is reclaimed area.
- **Therefore, the area required for the hydrogen plant can be accommodated in SCCL land**, even in the long term. However, land acquisition must be initiated with appropriate arrangement (e.g. leasing).

Production, lifetime, replacement.

Inputs	Units	Value
Hydrogen Production	Kg/yr	16,500,000
Electrolyser Stack Capacity	MW	80
Total Capex	USD/KW	670
O&M (% of Capex)	%/yr	2.50%
Energy Consumption	MWh/yr	841500
Electricity Cost	USD/MWh	30
Storage	USD/kg NH3	0.3
Period of Analysis	yrs	20
Discount Rate	%	5%

- For Elecctrolyzer and Ammonia Storage the lifetime is approximately **10 years.**
- Solar and Wind farm has a lifetime and depreciation(%) as **20 years and 12.5%.**
- Replacement cost for overall system is determined by the difference of total profit of system and the annual average depreciation.

Impacts and Benefits

RTC-RE model, Financial Model

Key benchmarks

i) SOEC Electrolyzer : -

Technical Maturity: Mature
Typical Footprint: Up to 0.5 m^2/kW
Purity: >99.999%
Start Response Time: <5 seconds to 1 mins
Shutdown Response Time: Few seconds
Load Range: 10-100%
Lifetime: >90,000 hours
Lower Heating Value: Possible
Power consumption: 37-43 kWh/Nm^3 H_2O at standard conditions
CAPEX: <670$/kW (installed)
OPEX: Yearly % of Capex: ⩽2%

ii) Ammonia Synloop:-

CAPEX: USD 120/kW/yr.
OPEX: 1.2-1.5% of EPC.

iii) Air Separation Unit:-

OPEX: ~3% of EPC

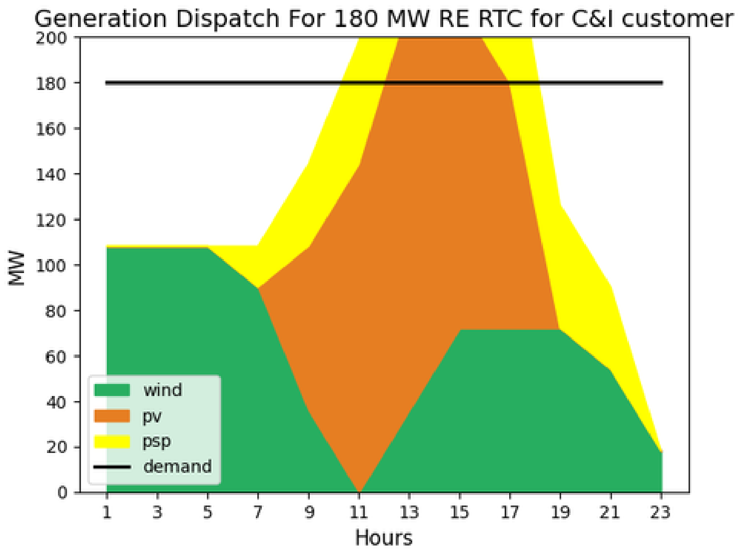
iv) Ammonia Storage

CAPEX: USD 950/ ton NH3.

Conversion Factors

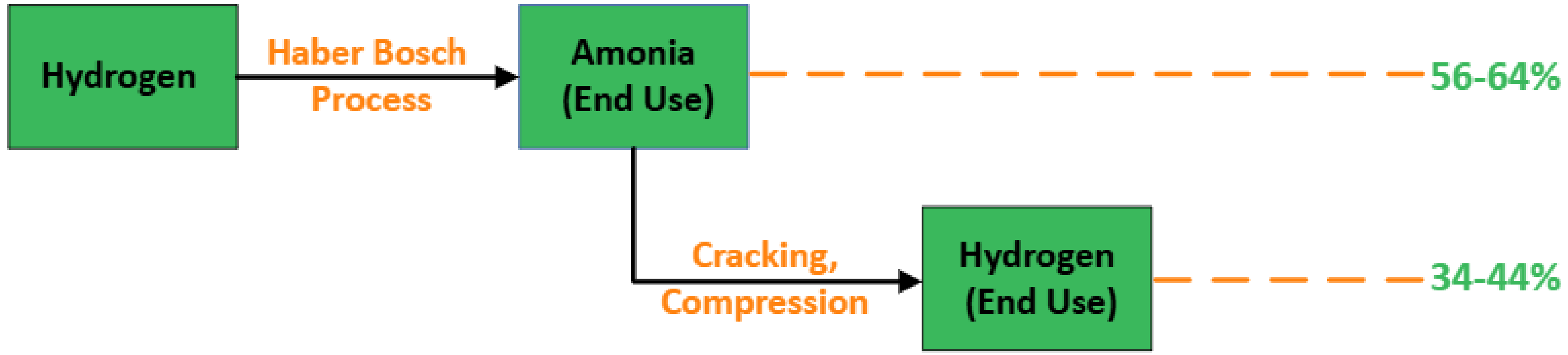
i) Energy to H2:
50-55 MWh energy per ton of H2.
***ii) Water to H2:* 9-10 m³ of water per ton of H2.**
***iii) H2 to NH3:* 5.5 tons of NH3 per ton of H2.**
iv) H2 generation per MW electrolyzer:
Approximately 100-150 tons per annum (TPA).

- The RTC-RE model for Ramagundum with 80MW load is plotted
- The CAPEX,OPEX and O&M costs alongside with the site constraints is considered
- The Techno-Economic factors that would've hindered the storage and production expenses are overcame.

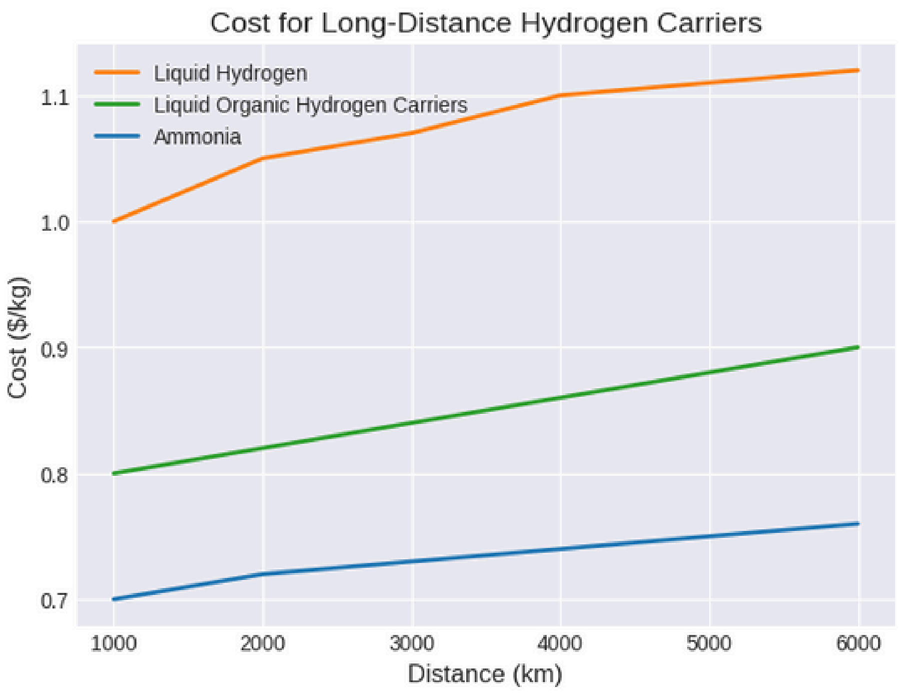


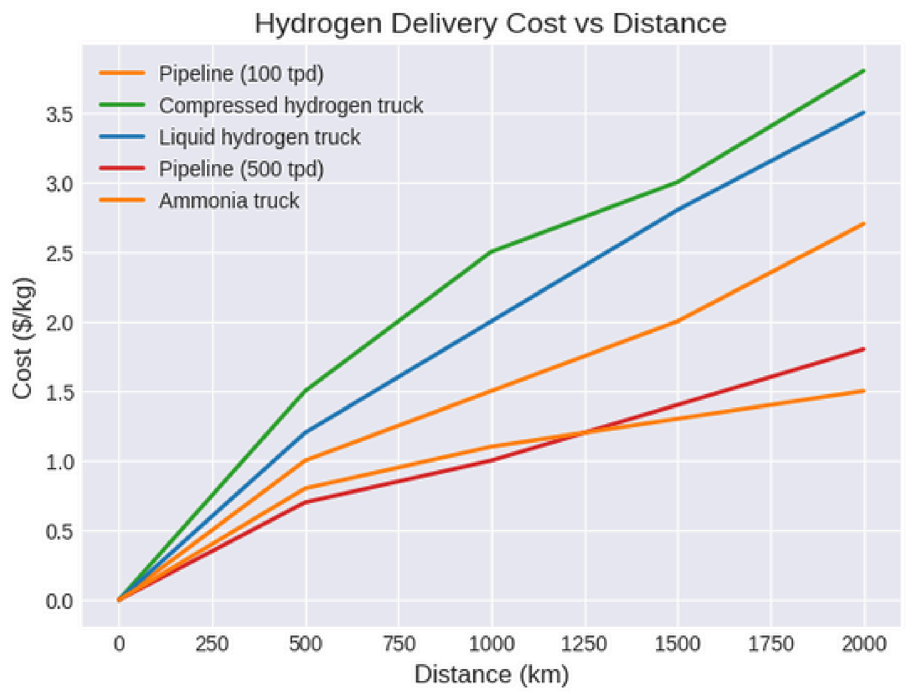
RTC-RE for 80MW Electrolyzer load

Storage technology



- Nearly 60TPD of green hydrogen is aimed to be produced from the electrolyzer units and using the **Haber Bosch Process** it is then proposed to be carried by ammonia and stored
- By using catalysis the green ammonia is again converted to green hydrogen and is then transported through high pressure **pipelines**





Ammonia; A better alternative for transport and storage

Future work, Research, and References

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Results

The cost of electricity (tariff) and the efficiency of Electrolyser are the major factors that affect the levelized cost of green hydrogen.

LCOH calculation :-

Parameter	Units	Cost/yr	NPV($)
O&M	USD	1,340,000	23,253,978
Electricty	USD	25,245,000	303,730,280
Storage	USD	5,100,000	71,145,000
Total	USD	**31,685,000**	**398,129,258**

Hydrogen production :
/yr = 16,500,000 kg
NPV* = 172,325,000 kg

Total Capex :
Cost/yr = 53,600,000 USD
NPV* = 53,600,000 USD

$$\textbf{LCOH} = \frac{\text{O\&M + Electricity Cost + Storage Cost + total Capex}}{\text{Net Present Value of hydrogen production}}$$

***Parameters in the above eq. holds NPV (Net present value)**

$$\textbf{LCOH} = \frac{(23253978+303730280+71145000+53600000)}{172325000}$$

LCOH(nearly) = 2$/kg

Future Work

- The hydrogen output from Solhyd and its CAPEX is to be researched upon.
- The policies and government incentives are to be taken into consideration for **economies of scale.**
- The hydrogen output from Solhyd and its CAPEX is to be researched upon.
- The pipelining for the transport and other facilities with the **safety** measures in-check is to be formulated.
- The **sizing and infrastructure optimization** of proposed green hydrogen facility is yet to be analyzed.

References

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